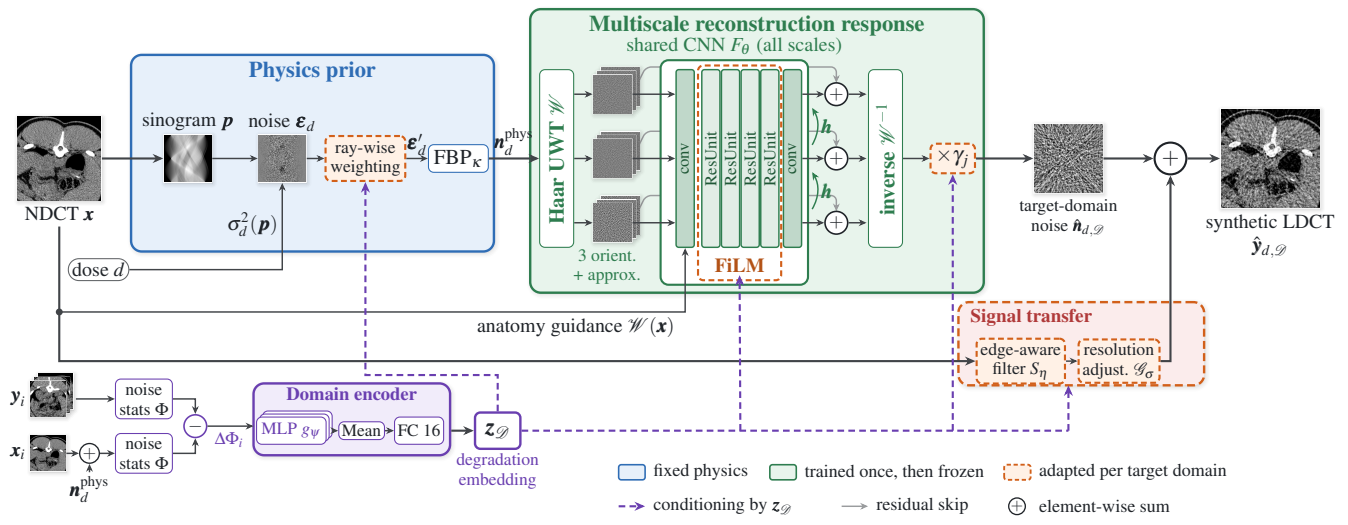

Physics prior
sinogram p
noise ε_d
ray-wise weighting
ε'_d
FBP$_\kappa$
n_d^{phys}
$\sigma_d^2(p)$
dose d
NDCT x
Multiscale reconstruction response
shared CNN F_θ (all scales)
Haar UWT $\mathscr{W}$
3 orient. + approx.
conv
ResUnit
ResUnit
ResUnit
ResUnit
conv
FiLM
h
h
inverse $\mathscr{W}^{-1}$
$\times\gamma_j$
target-domain noise $\hat{n}_{d,\mathscr{D}}$
synthetic LDCT $\hat{y}_{d,\mathscr{D}}$
anatomy guidance $\mathscr{W}(x)$
Signal transfer
edge-aware filter S_η
resolution adjust. $\mathscr{G}_\sigma$
y_i
x_i
noise stats Φ
noise stats Φ
n_d^{phys}
$\Delta\Phi_i$
Domain encoder
MLP g_ψ
Mean
FC 16
$z_\mathscr{D}$
degradation embedding
fixed physics
trained once, then frozen
adapted per target domain
conditioning by $z_\mathscr{D}$
residual skip
element-wise sum